

Calculus I Lesson 49

$$\frac{d}{dx}[\sinh^{-1} u] = \frac{1}{\sqrt{u^2 + 1}} \cdot u'$$

$$\frac{d}{dx}[\tanh^{-1} u] = \frac{1}{1 - u^2} \cdot u'$$

$$\frac{d}{dx}[\operatorname{sech}^{-1} u] = \frac{-1}{u\sqrt{1 - u^2}} \cdot u'$$

$$\frac{d}{dx}[\cosh^{-1} u] = \frac{1}{\sqrt{u^2 - 1}} \cdot u'$$

$$\frac{d}{dx}[\operatorname{coth}^{-1} u] = \frac{1}{1 - u^2} \cdot u'$$

$$\frac{d}{dx}[\operatorname{csch}^{-1} u] = \frac{-1}{|u|\sqrt{1 + u^2}} \cdot u'$$

Question 1:

$$f(x) = \cosh^{-1}(5x)$$

Find $f'(x)$

Hint: Let $u = 5x$

a) $\frac{du}{dx} = \underline{\hspace{2cm}}?$

b) $f'(x) = \underline{\hspace{2cm}}?$

Question 2:

$$y = 4x \cdot \tanh^{-1}(5x)$$

Find y' .

$$\text{Hint: } y' = 4x \cdot D_x(\tanh^{-1}5x) + \tanh^{-1}5x \cdot D_x(4x)$$

a) $D_x(\tanh^{-1}5x) = \underline{\hspace{2cm}}?$

b) $D_x(4x) = \underline{\hspace{2cm}}?$

c) $y' = \underline{\hspace{2cm}}?$

Question 3:

$$\text{Given: } D_x(\sinh^{-1}x) = \frac{1}{\sqrt{x^2+1}}$$

$$\int \frac{1}{\sqrt{x^2+1}} dx = \underline{\quad ? \quad} + C$$

Question 4:

Given: $D_x(\cosh^{-1}x) = \frac{1}{\sqrt{x^2-1}}$

$$\int \frac{1}{\sqrt{x^2-1}} dx = \underline{\quad ? \quad} + C$$

Question 5:

$$\text{Given: } D_x(\tanh^{-1}x) = \frac{1}{1-x^2}$$

$$\int \frac{1}{1-x^2} dx = \underline{\quad ? \quad} + C$$

